Yunkang Cheng | 程允康

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Department of Automation, Tsinghua University, Beijing, China

EDUCATION

Tsinghua University, Beijing, China **Aug. 2024 - 2029 (expected)**

B.Eng. in Automation (in progress), Department of Automation

- Cumulative GPA: 3.7/4.0.
- Selected coursework: Linear Algebra (A-), Calculus A(1) (A-), Object-Oriented Programming Practicum (A), Digital Electronics Laboratory (A-), Electronics Design Project (A-).

RESEARCH EXPERIENCE

Humanoid Roller-Skating with Adversarial Motion Priors **Dec. 2025 - Present**

Project Lead, Robot Control Laboratory, Tsinghua University | Advisor: Prof. Mingguo Zhao

Learning Roller-Skating Motions of Humanoid Robots Based on Adversarial Motion Priors

Yunkang Cheng*, *Yutong Wu**, *Menghan Li, Shihe Zhou, and Mingguo Zhao*

CLAWAR 2026, oral presentation. * Equal contribution.

arXiv:2607.10815 | Project website

- Lead a project on humanoid roller-skating with passive wheels, developing locomotion policies and coordinating simulation studies and hardware experiments.
- Trained separate Pump Glide and Push Glide policies in Isaac Lab using proximal policy optimization (PPO) with adversarial motion priors (AMP), retargeted human motion data, and domain randomization.
- Contributed to wheel-ground contact modeling and policy evaluation in MuJoCo; deployed policies on Booster T1. Each policy commands 21 body joints at 50 Hz while the skate wheels remain passive.
- Evaluated gait timing, velocity tracking, and posture in simulation; demonstrated both gaits on hardware and presented the work at CLAWAR 2026 in Cranfield, UK.
- Current work investigates skating initiation and transitions to gliding. Funding: National Innovation and Entrepreneurship Training Program for College Students; Tsinghua's Student Academic Research Advancement Program (Category B).

Humanoid Robot Soccer, Tsinghua Hephaestus **2025 - 2026**

Core Team Member | Advisor: Prof. Mingguo Zhao

- Developed obstacle-avoidance and navigation algorithms for humanoid soccer, including high-speed motion, field re-entry, and positioning during matches.
- Integrated and tested navigation behaviors with teammates during preparation and on-site deployment for RoboCup and the World Humanoid Robot Games.

SKILLS AND INTERESTS

Programming: Python, C/C++, MATLAB.

Tools and platforms: PyTorch, Isaac Lab, Isaac Gym, MuJoCo; Booster T1 and Qmini.

Research interests: Robot learning, reinforcement and imitation learning, humanoid locomotion, contact dynamics, and sim-to-real transfer.

SELECTED ENGINEERING PROJECTS

Passive Dynamic Walking: Modeling and Stability Analysis **Oct. 2026**

Course Project in Humanoid Robotics, Tsinghua University

- Modeled a passive biped with circular-arc feet in MATLAB using hybrid dynamics and impact maps; studied the effects of mass, center-of-mass location, and spring stiffness on gait.
- Computed fixed points of the Poincaré return map and assessed local stability using the spectral radius of its Jacobian; evaluated perturbation recovery over 50 steps for 48 parameter configurations.

Qmini Bipedal Robot: Assembly and Integration **Fall 2025**

Course Project in Manufacturing Engineering, Tsinghua University

- Assembled and modified a 12-DoF Qmini biped; resolved mechanical-fit and wiring issues, calibrated its standing posture, and tested motion-control functions.

HONORS AND AWARDS

Competition honors listed below were awarded to teams.

RoboCup, Humanoid Soccer League - Large Division **2026**

Champion, Tsinghua Hephaestus; Incheon, Republic of Korea.

World Humanoid Robot Games, 5v5 Soccer **2025, 2026**

Champion with Tsinghua Hephaestus; Large Robots category in 2026.

RoboCup China Open, Humanoid League (Large) **2026**

First Prize in soccer; Champion of the Technical Challenge.

Tsinghua Future Robotics Challenge **2026**

First Prize; Team Captain.

Second Embodied Intelligence Challenge, Department of Automation **2025**

First Prize; Deputy Team Captain.

National Encouragement Scholarship, Tsinghua University **2026**

Also received the Tsinghua University Comprehensive Excellence Award.

Spark Fellowship, Tsinghua University **2026**

Selected for the 20th cohort of Tsinghua's undergraduate research and innovation program.

Tang Chung Ying Moral Education Scholarship, Tsinghua University **2024**

LEADERSHIP AND SERVICE

Tang Chung Ying Volunteer Association, Tsinghua University **Feb. 2026 - Present**

Vice President for General Affairs (since May 2026); previously Head of General Affairs

- Oversee administrative coordination, financial and activity reporting, and liaison with the Cyrus Tang Foundation; allocate responsibilities and organize volunteer recruitment.

Embodied Intelligence Industry Field Study, Guangdong **Aug. 2026**

Team Leader, Department of Automation

- Led a 17-member team on an eight-day field study in Shenzhen and Guangzhou, examining embodied intelligence, unmanned aerial vehicles, and autonomous driving; coordinated visits, interviews, and reporting. The team received the department's Silver Award for social practice.

Rural Revitalization and Education Outreach, Fangzheng **Jan. 2026**

Team Leader, Tsinghua University

- Led a 15-member team in Fangzheng, Heilongjiang; organized robotics outreach and helped establish the Shuimu Yike online teaching program at Fangzheng No. 4 Primary School. The team received a School of Architecture Gold Award and a university-level Silver Award.